Judge Model Evaluation: Benchmarking Large Language Models for Clinical Quality Assessment in Patient Portal Messaging

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Abstract

The growing integration of Large Language Models (LLMs) in clinical communication raises fundamental questions about their reliability in high-stakes medical settings. While my previous research with Seegmiller et al. [12] addressed the challenges of LLM-based response generation and the associated clinician editing load, a critical secondary question remains: how reliably can these models independently evaluate clinical quality? In this study, I shift the focus from response generation to response evaluation, specifically investigating how well off-the-shelf foundation models perform as zero-shot clinical judges. By utilizing a curated dataset of 60 patient-provider interactions sourced from the r/AskDocs subreddit, I examine the capacity of four diverse judge models (Claude, OpenAI/GPT, Llama, and Qwen) to assess the clinical reasoning quality of 60 unique AI-generated replies against an expert-annotated ground truth. My results indicate that while larger, general-purpose models like GPT achieve higher accuracy, no model reached 100% agreement, revealing a persistent "medical gap" in models trained on general internet data. These findings highlight the limitations of current foundation models in clinical oversight and underscore that, in the absence of task-specific fine-tuning, AI should function as a clinical assistant rather than an autonomous decision-maker.

1 Introduction

The increasing adoption of AI-powered tools in healthcare has sparked both excitement and caution among clinicians, researchers, and patients alike. Patient portal messaging—a primary channel through which patients seek advice and follow-up care—generates a substantial administrative burden for healthcare providers [3]. Automating or assisting with these responses using LLMs offers a promising path toward reducing this burden while maintaining communication quality [7, 8].

However, a critical and largely unresolved challenge remains: *Can an autonomous "Judge Model" reliably evaluate and benchmark the clinical quality of AI-generated responses across*

diverse generative LLMs, replacing manual audit with an objective, scalable framework? Without a robust evaluation mechanism, deploying AI in clinical communication risks propagating inaccurate or unsafe medical advice at scale.

Recent work by Seegmiller et al. [12] has made significant strides toward understanding the alignment between LLM-generated response drafts and the responses that clinicians would actually send to patients, introducing the EditJudge framework and a clinically-validated thematic taxonomy of response elements. Their findings reveal that even the best-adapted LLMs reduce clinician editing load by only 25–26%, and that substantial epistemic uncertainty persists at the individual clinician level. A natural and important next question follows: rather than relying on carefully fine-tuned judge models as in their work, *can general-purpose foundation models serve as reliable zero-shot clinical evaluators?*

This research addresses that challenge by constructing a multi-model judge framework—benchmarking four state-of-the-art LLMs on their capacity to assess the clinical quality, safety, and accuracy of AI-generated patient portal responses. By comparing judge model outputs against a manually annotated dataset, we investigate the consistency and reliability of AI-driven clinical oversight and benchmark the current state of generative models in high-stakes medical communication.

2 Related Work

2.1 Patient Message Response Drafting and LLM Alignment

A growing body of work has examined the feasibility of using LLMs to draft responses to patient portal messages, with mixed results. Some studies report that LLM-generated drafts can be useful in promoting empathy and providing health advice [8, 9], while others find that these drafts frequently diverge from clinician responses in style, content, and accuracy [7, 10]. Importantly, divergence between LLM drafts and clinician responses raises dual concerns: *unreliability*, when drafts require significant clinician editing that defeats the purpose of AI-assisted efficiency, and *irresponsibility*, when low-quality drafts are sent to patients without adequate review [12].

Most directly related to the present work, Seegmiller et al. [12] proposed a comprehensive evaluation framework—the EditJudge framework—for assessing the *clinician editing load* imposed by LLM-drafted responses to patient portal messages. Their work made several key contributions that serve as the foundation for this study. First, they developed a clinician-validated taxonomy of eight thematic elements present in high-quality clinician responses: empathetic communication, symptom-related follow-up questions, medication-related follow-up questions, medical assessment, medical planning, logistics, care coordination, and contingency planning. Second, they introduced two complementary evaluation metrics: a *content-level edit-F1 score*, measuring the precision and recall of specific clinical facts and instructions in LLM drafts relative to expert-written responses, and a *theme-level edit-F1 score*, capturing higher-level thematic alignment between drafts and clinician intent. Third, they evaluated a range of LLM adaptation techniques—including thematic prompting, retrieval-augmented generation (RAG), supervised fine-tuning (SFT), and a novel thematic

direct preference optimization strategy (TADPOLE)—finding that contextual adaptation improves alignment by up to 33% over zero-shot baselines. Their best-performing adapted models achieved a content-level edit-F1 of approximately 0.25–0.26, suggesting that current LLMs would reduce clinician editing load by roughly one quarter.

Crucially, Seegmiller et al. also highlighted substantial epistemic uncertainty in aligning LLMs with individual clinicians: inter-annotator predictability (IAP) at the content level was only 0.24, meaning that even a different clinician's response serves as only a modest draft for another clinician. This finding underscores the need for LLM evaluation frameworks that go beyond response generation quality and directly assess the *judgment capabilities* of LLMs as evaluators of clinical content.

2.2 LLM-as-a-Judge Frameworks

The use of LLMs as automated judges of LLM-generated content has grown significantly in recent years [4, 11], including in healthcare text generation contexts [6]. Bavaresco et al. [4] conducted a large-scale empirical study on using LLMs as surrogates for human judges across 20 NLP evaluation tasks, finding both promise and significant caveats depending on model size and task domain. In the medical domain, benchmarks such as HealthBench [2] and MedHelm [6] systematically evaluate models on medical reasoning and communication tasks.

Most similar to the judge evaluation component of our work, the EditJudge framework introduced by Seegmiller et al. [12] employs a fine-tuned LLama3-8B-Instruct model as a content-level judge, achieving 96% agreement with expert human annotators. Their framework is, however, designed to evaluate *response drafting quality* under a specific fine-tuned judge—rather than to benchmark the *inherent judgment capabilities* of diverse foundation models in zero-shot clinical evaluation settings. Concurrent work has also examined AI responses in patient portal contexts more broadly. Baxter et al. [5] evaluated generative AI responses to patient messages in an institutional setting, while Biro et al. [7] surveyed the opportunities and risks of AI in patient portal messaging, highlighting both potential efficiency gains and the risks of clinically inaccurate responses.

2.3 The Present Work

This research extends the scope of Seegmiller et al. [12] by shifting the focus from *LLM response generation* to *LLM response evaluation*. Rather than asking how well LLMs can draft clinician responses, I investigate how effectively foundation models can independently evaluate and benchmark the clinical quality of such responses. To conduct this assessment, I independently curated and manually annotated a dataset of 60 patient-provider interactions from public forums. Using this as a ground-truth reference, I benchmark a diverse suite of off-the-shelf foundation models (Claude, OpenAI/GPT, Llama, and Qwen) in their capacity to serve as zero-shot clinical judges. By evaluating these public models without task-specific fine-tuning, I aim to determine whether general-purpose foundation models possess sufficient inherent clinical reasoning to augment or replace manual auditing in high-stakes medical communication.

3 Dataset Overview

3.1 Data Source and Inclusion Criteria

The dataset consists of 60 patient-provider interactions curated from the *r/AskDocs* subreddit. This platform serves as a public forum where users seek medical guidance and verified medical professionals provide responses. I selected interactions that satisfy the following inclusion criteria:

- **Doctor-verified responses:** Only interactions where the responding user is a verified medical professional were included.
- **Community validation:** Each included response had received a minimum of 10 upvotes, serving as a proxy for high-utility, community-validated clinical content.

3.2 Generation Pipeline

Three diverse foundation models served as the "Generator" layer: **Claude Sonnet 4**, **ChatGPT (GPT)**, and **Llama**. Each model was prompted to draft a response to one of the 60 patient portal messages under a **zero-shot protocol**—that is, without access to the original physician's response. This design choice ensures that model outputs reflect internal clinical reasoning rather than reliance on labeled training examples. The exact identifiers of the judge models used to evaluate these responses are listed in Appendix A.1.

The generation process yielded a total of **60 unique clinical responses** (20 per generator model), creating a high-variance dataset well-suited for rigorous downstream evaluation.

Two deliberate design choices shaped this layer. First, each patient message was answered by exactly one generator, so the 60 cases were partitioned into three disjoint sets of 20 rather than having all three models respond to all 60 messages. This avoids *near-duplicate responses* to the same prompt: if every generator answered every message, the judge models would repeatedly evaluate three minor variations of the same clinical content, and a single easy or hard case would be counted three times, inflating apparent agreement and entangling judge performance with the idiosyncrasies of a handful of prompts. Partitioning the messages instead spreads the evaluation across the full set of 60 distinct clinical scenarios, so each judgment contributes independent signal. Second, allocating 20 messages per generator keeps the generator mix balanced, ensuring that no single model's stylistic tendencies dominate the response pool and that downstream comparisons of judge behavior are not confounded by an uneven distribution of generator styles. Together these choices maximize the diversity of clinical content and writing styles the judges must contend with while holding the total annotation burden fixed at 60 cases.

3.3 Annotation Process

Rather than relying on existing datasets, I performed a manual annotation of these 60 interactions to establish a ground-truth reference. Each interaction was decomposed into individual sentence units, resulting in a total of **243 annotated sentence elements**. I

defined the clinical equivalence for each segment based on core medical intent—such as recommendations, diagnoses, and safety warnings—to ensure that the ground truth reflects the practical standard of clinical care.

For each sentence, I assessed whether the model's output conveyed the same clinical meaning as the expert-authored ground truth. To ensure rigorous evaluation, I adopted a strict binary match/no-match protocol. This approach specifically penalized over-generation by the LLMs; sentences produced by the generator that were not clinically represented in the initial ground-truth response were labeled as a "no-match," ensuring the evaluation captured only the precision of clinical reasoning rather than general verbosity. This manually curated dataset serves as the benchmark against which I evaluate the zero-shot reasoning of the foundation models.

4 Methodology

4.1 Judge Model Setup

Each judge model was tasked with reproducing the same sentence-level annotation procedure performed by the human annotator. The input to each judge was a CSV file in which every row contains the original patient *prompt*, the *doctor's response*, the full *model response* being evaluated, and a single *doctor-response sentence* to be matched. For each row, the judge must locate the sentence in the model response that conveys the same clinical meaning as that doctor sentence—or declare that no such sentence exists. The judge writes its decision to a new column, producing an output CSV identical to the input but with an added *matched_sentence* field holding either the model sentence it selected or a *no-match* marker.

Concretely, each of the 60 patient-provider interactions had its doctor response decomposed into individual sentence elements, yielding 243 ground-truth sentences in total. Each ground-truth sentence was paired with the corresponding full model response to form one row, producing 243 (ground-truth, model-response) evaluation units that were held identical across all four judge models. Because the doctor's response is decomposed this way (one row per doctor sentence), iterating over all rows reproduces the full human annotation: each doctor sentence is either paired with a matching model sentence or left unmatched. These per-row decisions are exactly what is later scored, so the evaluation measures how well each model performs the matching task at the sentence level.

The matching criteria (the "Match Definition") were defined as follows:

1. **Clinical Equivalence:** Matches are determined by core clinical intent—including recommendations, diagnoses, safety warnings, and prognoses—rather than verbatim wording.
2. **Acceptable Variation:** Paraphrasing and structural differences are permitted, provided the underlying medical guidance is identical.
3. **Exclusions:** Partially overlapping statements, topically related but clinically distinct advice, and vague or non-specific generalities are strictly marked as non-matches.

The four judge models used were: **Llama**, **OpenAI (GPT)**, **Qwen**, and **Claude**. This selection spans both open-source and commercial model families, enabling a broad characterization of zero-shot judgment capabilities across diverse architectures. The exact model identifiers are listed in Appendix A.1.

4.2 Scoring

Each row of judge output was scored against a manual gold standard using a binary protocol:

- **Score of 1:** The judge's call was correct—either a claimed match in which the selected sentence genuinely conveyed the same core clinical intent as the doctor sentence, or a correct abstention when no valid equivalent existed in the model response.
- **Score of 0:** The judge's call was incorrect—either a *forced match* (the selected sentence was clinically distinct, only partially overlapping, or vague), or a *missed match* (a valid clinical equivalent existed but the judge returned no-match).

Accuracy is reported as the number of rows scored 1 divided by the total number of rows. Because the protocol treats every row independently and applies the same gold standard across all four judges, the resulting scores are directly comparable.

I adopted this binary semantic-matching approach because traditional string-matching proved insufficient for clinical evaluation. Generative models frequently use diverse phrasing to convey the same clinical intent, so exact string-matching would unfairly penalize minor lexical or stylistic variation. Evaluating for clinical equivalence instead ensures that the score reflects the model's underlying reasoning accuracy rather than its adherence to a specific vocabulary, providing a clear, interpretable metric for clinical alignment and a fair comparison across judges.

5 Results

5.1 Overall Accuracy

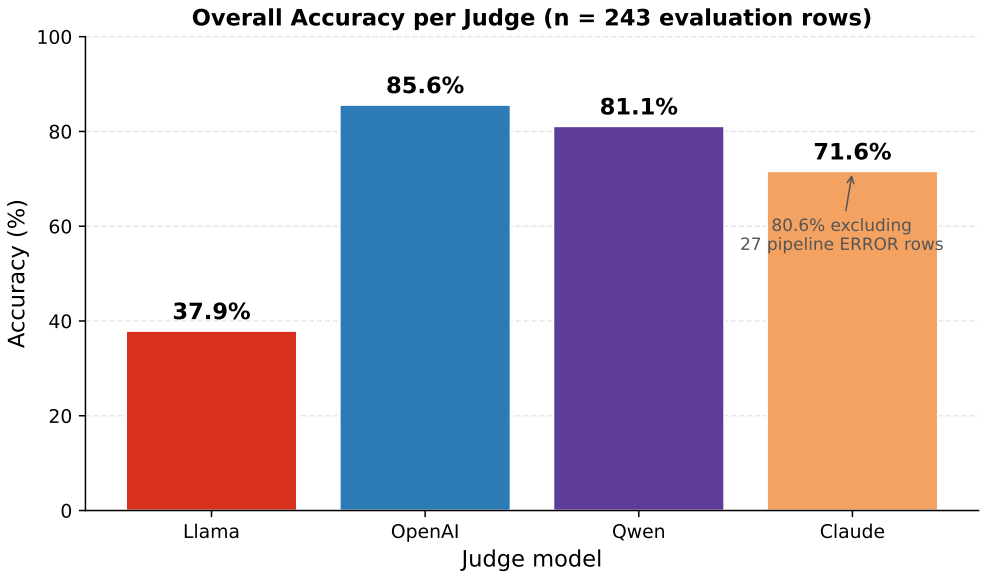


Figure 1: Overall accuracy of each judge on the 243-row evaluation set. Llama trails the other three judges by a wide margin (37.9%). OpenAI is the strongest (85.6%), with Qwen close behind (81.1%). Claude's headline figure of 71.6% is depressed by 27 rows on which the judging tool returned an error string instead of a verdict; on the 216 rows where Claude produced a verdict, its accuracy is 80.6%.

Across the 243 evaluation rows, OpenAI achieves the highest overall accuracy at 85.6%, followed by Qwen at 81.1%, Claude at 71.6%, and Llama at 37.9%. The Claude figure deserves an immediate caveat: 27 of its 243 rows (11.1%) returned the literal string "ERROR" as the matched sentence, indicating a pipeline failure rather than a content judgment. These errors arose during inference through the Langchain Dartmouth interface. The failures initially appeared to be token-limit issues, but they persisted even after token limits were raised; the 27 errors reported here represent the lowest error count achievable for Claude across repeated runs, and their precise cause could not be isolated. The failures were also not randomly scattered—they fall on 27 consecutive rows (214–242) that continue unbroken to the end of the file, several of them structurally simple cases the other three judges handled cleanly, which further indicates a tooling failure rather than a difficulty with the underlying clinical content.

Two Claude figures must therefore be distinguished: **71.6%** if the ERROR rows count as incorrect, and **80.6%** (174 / 216) on the rows where Claude produced a verdict. The former measures the *operational reliability* of the pipeline as deployed; the latter measures Claude's *judgment quality* on the rows it actually evaluated.

This single-metric view establishes a coarse ordering—Llama is notably weaker than the other three, and OpenAI, Qwen, and Claude (on rows it evaluated) cluster within five percentage points of one another. But accuracy alone hides a critical fact about these judges: they differ not so much in how often they are right, but in *what kinds of errors they make*. The

remainder of this section examines those differences directly.

5.2 Beyond Accuracy: Confusion Matrix Analysis

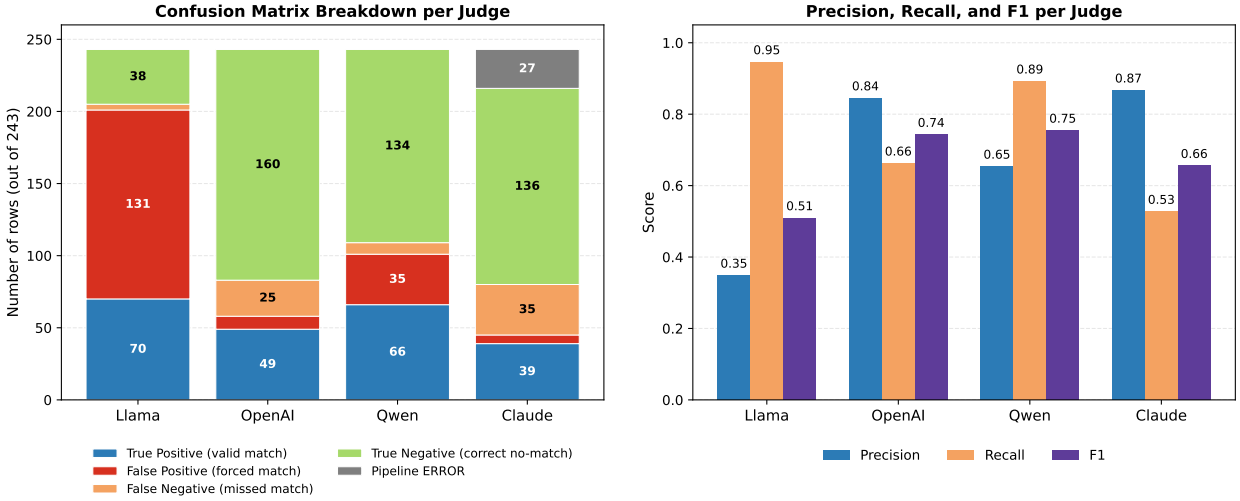


Figure 2: Confusion matrix breakdown (left) and precision/recall/F1 (right) per judge. The left panel shows how each judge's 243 rows distribute across true positives (claimed a match and a valid match existed), false positives (claimed a match when none existed), true negatives (correctly abstained), false negatives (missed a real match), and pipeline errors. The right panel translates these counts into precision, recall, and F1.

The error distribution becomes interpretable once each row is classified by *what kind* of correct or incorrect outcome it represents. Treating a judge's task as a binary detection problem—predict whether any valid clinical equivalent exists in the model response—each row falls into one of four cells: a *true positive* (judge claimed a match in a row where a valid equivalent exists), a *false positive* (judge claimed a match in a row where none exists), a *true negative* (judge correctly abstained), or a *false negative* (judge abstained when a valid equivalent did exist). Crucially, whether a row actually contains a valid match is a fixed property of the (ground-truth sentence, model-response) pair—it does not depend on which judge is evaluating it. Either the model response contains a clinically equivalent sentence or it does not. The judges differ only in whether they correctly find these matches, not in how many exist. In this dataset, 74 of the 243 rows contain at least one valid clinical equivalent and 169 do not; these 74 rows are the same for every judge.

Three standard metrics are then derived: **precision** (of all matches the judge claimed, what fraction were in rows where a valid equivalent actually exists), **recall** (of all rows where a valid equivalent exists, what fraction the judge correctly flagged as a match), and their harmonic mean, **F1**. Because a judge can claim a match but pick the *wrong sentence* even in a row where a valid equivalent exists, an additional column reports **sentence-finding accuracy**: of the matches the judge claimed, what fraction selected the clinically correct sentence under the strict criteria. The two views (match-existence vs. sentence-finding) measure different things—detecting *that* clinical agreement exists vs. locating *where* it lives—and the gap between them turns out to be diagnostic for one of the judges.

Table 1: Per-judge confusion matrix and derived metrics.

Judge	TP	FP	TN	FN	ERR	Precision	Recall	F1	Sent-acc
Llama	70	131	38	4	0	0.35	0.95	0.51	54 / 201 = 0.27
OpenAI	49	9	160	25	0	0.85	0.66	0.74	47 / 58 = 0.81
Qwen	66	35	134	8	0	0.65	0.89	0.75	63 / 101 = 0.62
Claude	39	6	136	35	27	0.87	0.53	0.66	37 / 45 = 0.82

Note: Pipeline ERROR counts rows where the Claude judging tool returned a literal "ERROR" string instead of a verdict; these 27 rows are excluded from Claude's confusion matrix (denominator = 216). *Sent-acc* (sentence-finding accuracy) reports, of the matches the judge claimed, the fraction in which the specific sentence selected was a valid clinical equivalent under the strict criteria.

Four observations follow directly from Table 1 and Figure 2, and substantially complicate the ordering implied by Figure 1:

1. **Llama is not weak in the same sense that the other judges are weak; it is dominated by a single failure mode.** With recall of 0.95, Llama detects almost every row in which a valid clinical equivalent exists—its false-negative count is only 4 out of 74 positive-class rows. Its problem is precision (0.35): roughly two out of every three matches it claims correspond to rows in which no valid equivalent exists at all. Llama's aggregate accuracy of 37.9% is therefore not a measure of inability to recognize clinical agreement; it is a measure of overcommitment to recognizing it.

2. **Llama's sentence-finding accuracy collapses to 0.27**, roughly a third of its precision under the detection framing. The gap between Llama's match-existence behavior (catches most real matches) and its sentence-finding behavior (picks the wrong sentence three out of four times when committing to a row) quantifies a specific failure mode: when Llama commits to a row, it tends to grab any thematically related sentence rather than the clinically equivalent one. The other three judges show much smaller gaps between their precision and their sentence-finding accuracy (0.81 / 0.85 for OpenAI, 0.62 / 0.65 for Qwen, 0.82 / 0.87 for Claude), indicating that when they commit, they tend to commit to the right sentence.

3. **OpenAI and Qwen achieve nearly identical F1 (0.74 vs. 0.75) through opposite mechanisms.** OpenAI is precision-favored (precision 0.85, recall 0.66); Qwen is recall-favored (precision 0.65, recall 0.89). This is consistent with their abstention rates: OpenAI abstained on 76% of rows, Qwen on 58%, and the two judges concentrate their errors on different sides of the precision/recall trade-off. The two also differ on sentence-finding accuracy (0.81 vs. 0.62), suggesting OpenAI is the more reliable choice when a downstream task requires the exact matching sentence rather than just a match-existence verdict.

4. **Claude has the highest precision of the four (0.87) and the highest sentence-finding accuracy (0.82)**, but the lowest recall (0.53), missing roughly half of the valid matches that exist in the dataset. Its headline accuracy in Figure 1 is depressed both by this conservatism and by the pipeline reliability issue.

5.3 Hallucination Rate

A *hallucinated match* is defined as a claimed match whose text does not occur as a substring of the corresponding model response, even after case-folding and substring tolerance for minor formatting differences. The results are summarized in Table 2.

Table 2: Match provenance and pipeline reliability per judge.

Judge	Matches claimed	Hallucinations	Hallucination rate	Pipeline ERRORs
Llama	201	7	3.5%	0
Qwen	101	1	1.0%	0
Claude	45	0	0.0%	27
OpenAI	58	0	0.0%	0

Note: Hallucination rate is the fraction of a judge's claimed matches whose text does not appear in the model response being evaluated. *Pipeline ERRORs* count rows where the judging tool returned a literal "ERROR" string instead of a verdict (see Section 5.1). Hallucination counts are computed only over rows where the judge produced a verdict.

Across the eight observed hallucinations (seven from Llama, one from Qwen), the fabricated text followed one of three patterns: an exact verbatim copy of the doctor's ground-truth sentence presented as if the model had written it; a synthetic sentence that appeared nowhere in the model response; or a concatenation of separate items stitched into a single sentence. Worked examples of each pattern are provided in Appendix A (Section A.7). The underlying behavior is consistent: when no valid match exists in the model response but the judge has committed to producing one anyway, the more aggressive judges occasionally manufacture text rather than abstain. This is the "force associations" behavior described qualitatively in earlier drafts, now quantified, and it tracks judge aggressiveness directly—Llama, which claims the most matches, also fabricates the most.

6 Discussion

6.1 Implications for Clinical AI Deployment

The results carry important implications for the deployment of AI systems in clinical communication. The finding that no model achieved perfect accuracy reinforces the position that AI tools should currently serve as assistants or "second opinions" rather than autonomous decision-makers in healthcare. Even the best-performing judge missed clinically meaningful distinctions in a non-trivial fraction of cases—a failure mode with potentially serious consequences in high-stakes medical contexts.

Most concerning is the tendency of these models to force associations: when faced with content they cannot verify, they often claim a match rather than identifying a lack of clinical equivalence. This took several recurring forms. The most aggressive judges reused a single loosely related model sentence as the claimed match for multiple distinct ground-truth elements, and in the clearest failures fabricated a match outright—returning a sentence that never appeared in the model response, sometimes by copying the doctor's ground-truth sentence verbatim. More troubling clinically, judges across all families sometimes matched a

doctor sentence to a model sentence on the same topic but of directly opposing stance—for instance, a statement respecting patient autonomy matched to a recommendation to override the patient's wishes—indicating that the matching criterion rewards topical relatedness without verifying directional agreement. In a medical setting this is especially problematic, as it presents inaccurate or even contradictory information as a verified match. Worked examples of each pattern, with full patient context, are provided in Appendix A.

While these models demonstrate utility as drafting or support tools, these findings suggest that the current generation of LLMs—whether used to generate or to evaluate clinical responses—falls short of the reliability threshold required for autonomous deployment in patient care.

6.2 Zero-Shot Judges vs. Fine-Tuned Judges

A key distinction between our work and the EditJudge framework [12] is the reliance on zero-shot foundation models rather than task-specifically fine-tuned judges. Seegmiller et al. demonstrated that fine-tuning a LLama3-8B model on only 450 annotated samples was sufficient to achieve 96% agreement with expert human annotators on content-level matching decisions. Our results suggest that zero-shot foundation models, even large commercial ones, fall considerably short of this benchmark—highlighting the value of domain-specific supervision for clinical judgment tasks, even when model scale is large.

6.3 The Role of Restraint vs. Aggressiveness

The underperformance of Llama highlights an important principle: judging behavior is shaped not only by model scale but by how aggressively a model commits to a match in the face of ambiguity. The precision/recall analysis in Section 5.2 shows that Llama's failure is not an absence of clinical recognition—its recall is the highest of the four judges—but an absence of restraint. The other three judges, all of which abstained more readily, achieved substantially better precision; the trade-off between sensitivity and specificity is therefore the central axis along which these judges differ. The conservative judges' characteristic error is the mirror image of Llama's: rather than forcing false matches, they abstain when the doctor's casual phrasing and the model's formal phrasing express the same clinical content in different registers, missing genuine matches.

It should be noted, however, that model scale is a plausible confounder in this comparison and cannot be fully separated from judging behavior with the present data. As detailed in Appendix A.1, the Llama judge is an 11B-parameter model, whereas the other three judges are substantially larger—the GPT judge has roughly 120B parameters and the Qwen judge 32B. Llama's low restraint may therefore reflect not only a distinct behavioral disposition but also the reduced capacity of a much smaller model to reason about fine-grained clinical equivalence under the strict matching criteria. Disentangling these two factors would require evaluating judges of comparable scale across families, which we leave to future work. The qualitative point—that aggressiveness, not raw recognition, drives Llama's aggregate error rate—holds regardless; but the magnitude of its underperformance should not be read as a pure measure of family-level capability given the scale asymmetry.

6.4 General Training Data vs. Medical Expertise

The persistent accuracy gap across all models is consistent with the well-documented challenge of applying general-purpose LLMs to specialized domains. Clinical reasoning requires not only broad language understanding but also a deep familiarity with medical terminology, clinical protocols, differential-diagnosis logic, and patient-safety considerations—knowledge that is underrepresented in standard internet-scale training.

7 Conclusion

This research investigated the capability of off-the-shelf foundation models to function as zero-shot clinical evaluators. By benchmarking four diverse judge models against a manually annotated dataset of 243 sentence-level evaluation units, I have demonstrated that while large-scale models show significant promise in clinical reasoning, they are not yet interchangeable with expert human audit. The precision/recall analysis revealed that the judges differ less in their overall ability and more in how they distribute their errors: Llama trades restraint for sensitivity, OpenAI trades sensitivity for restraint, Qwen sits between the two, and Claude achieves the highest precision at the cost of the lowest recall. A reproducible provenance check further established that the more aggressive judges occasionally fabricate matches outright, with Llama hallucinating 3.5% of its claimed matches and Qwen 1.0%. These results suggest that while foundation models can provide valuable assistance, they remain limited by a persistent "medical gap," confirming that expert human oversight is currently non-negotiable for high-stakes medical communication.

8 Acknowledgements and Limitations

I would like to acknowledge that the dataset used in this study was sourced from *r/AskDocs*. While these interactions are community-validated, they do not necessarily reflect the institutional standards of clinical care found in electronic health records; consequently, the "ground-truth" responses may contain variations in style or clinical depth. Furthermore, I performed all manual annotations independently, and I am not a medical professional.

Additionally, the binary scoring protocol—while chosen for its objectivity and clarity—may impose a harsh penalty on generative models. In instances where a model produced a response that was partially correct but semantically incomplete, or where the clinical intent was largely aligned but not identical to the doctor's response, the model received a score of zero. This strict criterion was selected to prioritize clinical safety and precision, but it should be noted that this approach may underestimate the utility of these models in less stringent, supportive workflows.

In addition to the above, the following methodological gaps should be made explicit:

- The gold-standard correctness labels were produced by a single non-clinician annotator. Inter-annotator agreement and physician review are required before these labels should be cited as a benchmark.

- The Claude judging tool produced a pipeline error on 27 of 243 rows (11.1%). These rows were treated as incorrect for the headline accuracy figure but should be re-run on a stable pipeline before the Claude numbers are cited as a substantive comparison against the other judges.
- The judge pool spans a wide range of parameter scales (an 11B Llama judge against a 120B GPT judge and a 32B Qwen judge; see Appendix A.1). Cross-family comparisons of judging behavior are therefore partially confounded by scale, and the magnitude of Llama's underperformance in particular should not be attributed to family-level capability alone.
- The benchmark is drawn from community-validated forum responses, not from electronic health records. The reported accuracy/precision/recall figures therefore measure agreement with verified-Reddit-doctor content rather than agreement with institutional clinical-documentation standards. This distinction should be reflected in how the results are interpreted in downstream work.

9 Future Work

The primary direction for future work involves domain-specific fine-tuning to close the observed performance gap. Future iterations of these models should be trained on specialized medical data—to better internalize the logic and rigorous conventions required for accurate clinical reasoning.

Additional avenues for independent investigation include:

- **Expanding the evaluation scope:** Increasing the dataset size beyond Reddit-sourced interactions to include diverse, clinically validated sources, which would help generalize findings across different patient populations and communication styles.
- **Confidence-based calibration:** Moving beyond binary scoring to evaluate how well model confidence scores align with actual decision accuracy, providing a clearer characterization of uncertainty in clinical AI evaluation.
- **Human-in-the-loop validation:** Conducting expert-led studies to compare model judgments against a consensus panel of physicians, establishing a more robust gold-standard for evaluating clinical AI reasoning.

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A Appendix

A.1 Model Versions

The exact model identifiers used in this study are listed in Table 3, for reproducibility. All models were accessed through the Langchain Dartmouth interface.

Table 3: Exact judge model identifiers.

Role	Family	Identifier
Judge	Llama	`meta.llama-3.2-11b-vision-instruct`
Judge	OpenAI/GPT	`openai.gpt-oss-120b`
Judge	Qwen	`qwen.qwen3-vl-32b-instruct-fp8`
Judge	Claude	`anthropic.claude-sonnet-4-20250514`

Note the substantial disparity in parameter scale across the judge pool: the Llama judge is an 11B-parameter model, whereas the GPT (`gpt-oss-120b` [1]) and Qwen (32B) judges are considerably larger. This asymmetry should be kept in mind when interpreting the cross-model comparisons in Section 5.2, and is discussed further in the Discussion (Section 6.3) and Limitations (Section 8).

A.2 The Judge Prompt

Each judge received the following system prompt, applied once per row (i.e., once for each *Doctor Sentence*, *Model Response* pair):

```
You are an expert medical data analyst.
You will receive one 'Doctor Sentence' and one 'Model Response'.
Find the sentence within the 'Model Response' that matches the clinical meaning
of the 'Doctor Sentence'.

RULES:
- If a match exists, return ONLY: MATCH: [exact sentence from Model Response]
- If no match exists, return:  NO MATCH
- Do not provide explanations or chat.
```

A.3 True Positive: Correct Match Identification

A claimed match in which the selected sentence is a valid clinical equivalent of the ground-truth sentence under the strict criteria of Section 4.1.

Context. A 19-year-old unvaccinated woman living with her unvaccinated mother asks whether getting vaccinated could expose her mother to COVID. *(OpenAI judge.)*

Ground truth. "The vaccine does not put COVID in your body."

Model excerpt. "You cannot transmit COVID-19 to your mother from receiving any of the approved COVID vaccines—they do not contain live virus and cannot cause infection or make you contagious."

Classification. True positive. The model and the doctor make the same scientific claim through different wording: COVID vaccines do not introduce live virus into the body.

A.4 True Negative: Correct Abstention

A correct *no match*, returned because no sentence in the model response is a valid clinical equivalent of the ground-truth sentence.

Context. The Reddit doctor opens the response with a self-disclosure. *(OpenAI judge.)*

Ground truth. "Not a medical doctor."

Model excerpt. None—the AI-generated response contains no analogous credentialing statement.

Classification. True negative. The doctor's credentialing disclaimer is an artifact of the forum context and is correctly absent from the AI draft; abstaining is the right call.

A.5 False Positive: Forced Match

A claimed match where no valid clinical equivalent existed—here, two sentences of directly opposing stance.

Context. A young adult asks how to convince their father to seek evaluation for symptoms suggestive of cancer. *(Llama judge.)*

Ground truth. "If he doesn't want to get checked out, you can't make him."

Llama's claimed match. "Please call emergency services or take him to an ER immediately; if he refuses, consider contacting adult protective services, as his deteriorating mental state may indicate he's no longer capable of making safe decisions for himself."

Classification. False positive, and a particularly concerning one. The doctor's sentence acknowledges patient autonomy; the model's recommends mechanisms to override the patient's refusal. The two stances are not merely different—they are opposed.

A.6 False Negative: Missed Match

A *no match* returned even though a valid clinical equivalent existed—most often when the same content appears in a different register.

Context. A teenager describes a recent acetaminophen overdose and asks whether they should have gone to the hospital. *(Claude judge.)*

Ground truth. "Yes, you absolutely should have gone to the hospital."

Available match in model. "You need to see a doctor NOW for bloodwork to check your liver function—acetaminophen overdose can cause serious liver damage that has no symptoms until it's severe."

Classification. False negative. The model conveys the same urgency about post-overdose evaluation, but the judge did not recognize the equivalence between the doctor's past-tense framing ("should have gone") and the model's present-tense recommendation ("need to see a doctor NOW").

A.7 Hallucinated Match

A claimed match whose text does not appear in the model response at all, verified by case-insensitive substring search.

Context. A patient describes constipation symptoms after starting Prozac. *(Qwen judge.)*

Ground truth. "Prozac does not treat constipation."

Qwen's claimed match. "Prozac does not treat constipation."

Actual model response. Discusses possible gastrointestinal causes of the symptoms but never states that "Prozac does not treat constipation."

Classification. Hallucination. The judge satisfied the surface form of the task—supplying a sentence that matches the ground truth—by copying the ground-truth sentence verbatim rather than locating genuine content in the model response.

A.8 Pipeline Error

A row on which the Claude judging tool returned the literal string "ERROR" instead of a verdict (see Section 5.1).

Context. A patient describes facial symptoms; the doctor follows up with a clarifying question. *(Claude judge.)*

Ground truth. "Can you still move the muscles in your face when this happens?"

Claude's output. `ERROR`

Classification. Pipeline error. The row is structurally similar to many the Claude judge handled correctly elsewhere; the verbatim "ERROR" string reflects a tooling failure, not a content judgment.